LP Duality – Lower Bounds – IP (Lec12)

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Todays topics & Logistics

- Duality in LP
  - Derivation of the Dual LP
  - Weak and Strong Duality

Duality in LP

- **Motivational Example:** consider the LP (from Lec12.pdf) → → →

$$\begin{aligned} max \quad & f(x) = x_1 + 6x_2 \\ & x_1 \le 200 \qquad (1) \\ s.t. \quad & x_2 \le 300 \qquad (2) \\ & x_1 + x_2 \le 400 \qquad (3) \\ & x_1, x_2 \ge 0 \end{aligned}$$

- Your instructor tells you that $\mathrm{x}^* = (x_1^*, x_2^*)^T = (100,300)^T$ is the optimal solution, with $f(x^*) = 1900$.

- How can you fact-check him/her ??

  - x^* is feasible

  - x_2^* is at its UB

  - x_1^* can still increase to reach its UB, but it will violate (3) unless we decrease x_2^*

Duality in LP

- **Motivational Example (cont.):**
  - How about we combine constraints (1) and (2) to produce a new constraint (**valid inequality**) ??
  - This constraint should produce an upper bound on the objective function
  - How about we **multiply (2) by 6 and add it to (1):**

$$x_1 + 6x_2 \leq 200 + 6 \times 300 = 2000$$

  - We have formed the objective at the LHS and bounded it (from above) by 2000

$$\begin{array}{lll} max & f(x) = x_1 + 6x_2 & \\ & x_1 \leq 200 & (1) \\ s.t. & x_2 \leq 300 & (2) \\ & x_1 + x_2 \leq 400 & (3) \\ & x_1, x_2 \geq 0 & \end{array}$$

Duality in LP

- **Motivational Example (cont.):**
  - Can we do better ? → find a smaller UB than 2000?

$$
\begin{aligned}
max \quad & f(x) = x_1 + 6x_2 \\
& x_1 \leq 200 \qquad (1) \\
s.t. \quad & x_2 \leq 300 \qquad (2) \\
& x_1 + x_2 \leq 400 \quad (3) \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Duality in LP

$$\begin{aligned} max \quad & f(x) = x_1 + 6x_2 \\ & x_1 \le 200 \qquad (1) \\ s.t. \quad & x_2 \le 300 \qquad (2) \\ & x_1 + x_2 \le 400 \quad (3) \\ & x_1, x_2 \ge 0 \end{aligned}$$

- **Motivational Example (cont.):**
  - Can we do better ? → find a smaller UB than 2000?
  - How about we **multiply (2) by 5 and add it to (3)**

$$5x_2 + (x_1 + x_2) \le 5 \times 300 + 400 \Rightarrow x_1 + 6x_2 \le 1900$$

  - **WOW**! We just showed that the objective function of the LP cannot exceed 1900
  - And your instructor did not make a mistake/typo (this time…)

Duality in LP

- Let's **formalize** what we did in the previous slides
  - We were looking for specific numbers to **multiply** the constraints and then **add** them together to form a **<u>valid inequality</u>** that provide an UB to the objective of the LP.

$$y_1x_1 + y_2x_2 + y_3(x_1 + x_2) \leq 200y_1 + 300y_2 + 400y_3 \quad \Rightarrow$$

$$\Rightarrow \underbrace{(y_1 + y_3)x_1 + (y_2 + y_3)x_2}_{\textbf{LHS}} \leq \underbrace{200y_1 + 300y_2 + 400y_3}_{\textbf{RHS}}$$

- The **LHS** looks like the objective function and hence we want its coefficients to satisfy

$$y_1 + y_3 \geq 1 \quad \text{and} \quad y_2 + y_3 \geq 6$$

- Also, we want **the best possible UP** and hence we minimize the **RHS**

$$\mathbf{min}\ \mathbf{200y_1 + 300y_2 + 400y_3}$$

$$\begin{array}{lll} max & f(x) = x_1 + 6x_2 & \\ & x_1 \leq 200 & (1) \\ s.t. & x_2 \leq 300 & (2) \\ & x_1 + x_2 \leq 400 & (3) \\ & x_1, x_2 \geq 0 & \end{array}$$

constraints		***multipliers***
$x_1 \leq 200$	(1)	y_1
$x_2 \leq 300$	(2)	y_2
$x_1 + x_2 \leq 400$	(3)	y_3

Duality in LP

- Let's formalize what we did in the previous slides

$$\begin{array}{lll} max & f(x) = x_1 + 6x_2 & \\ & x_1 \leq 200 & (1) \\ s.t. & x_2 \leq 300 & (2) \\ & x_1 + x_2 \leq 400 & (3) \\ & x_1, x_2 \geq 0 & \end{array}$$

  - We were looking for specific numbers to multiply the constraints and then add them together to form a valid inequality that provides an UB to the objective of the LP.

$$y_1 x_1 + y_2 x_2 + y_3(x_1 + x_2) \leq 200y_1 + 300y_2 + 400y_3 \quad \Rightarrow$$

$$\Rightarrow \underbrace{(y_1 + y_3)x_1 + (y_2 + y_3)x_2}_{\textbf{LHS}} \leq \underbrace{200y_1 + 300y_2 + 400y_3}_{\textbf{RHS}}$$

constraints		***mult.s***
$x_1 \leq 200$	(1)	y_1
$x_2 \leq 300$	(2)	y_2
$x_1 + x_2 \leq 400$	(3)	y_3

- The **LHS** looks like the objective function and hence we want its coefficients to satisfy

$$y_1 + y_3 \geq 1 \quad \text{and} \quad y_2 + y_3 \geq 6$$

- Also, we want the best possible UP and hence we minimize the **RHS**

$$\boldsymbol{min \;\; 200y_1 + 300y_2 + 400y_3}$$

$$\begin{array}{ll} min & g(y) = 200y_1 + 300y_2 + 400y_3 \\ s.t. & y_1 + y_3 \geq 1 \\ & y_2 + y_3 \geq 6 \\ & y_1, y_2, y_3 \geq 0 \end{array} \qquad \begin{array}{l} y^* = (0,5,1) \\ g(y^*) = 1900 \end{array}$$

Duality in LP

- We have focused on two optimization problems
  - The original problem is called the **Primal problem**
    - It is a **max** problem
  - The one we derived is called the **Dual problem**
    - It is a **min** problem
    - It provides an **Upper Bound** to the optimal objective of the Primal problem

Primal problem

$$
\begin{aligned}
max \quad & f(x) = x_1 + 6x_2 \\
& x_1 \leq 200 \qquad (1) \\
s.t. \quad & x_2 \leq 300 \qquad (2) \\
& x_1 + x_2 \leq 400 \quad (3) \\
& x_1, x_2 \geq 0
\end{aligned}
$$

$$x^* = (100,300)^T$$
$$f(x^*) = 1900$$

Dual problem

$$
\begin{aligned}
min \quad & g(y) = 200y_1 + 300y_2 + 400y_3 \\
s.t. \quad & y_1 + y_3 \geq 1 \\
& y_2 + y_3 \geq 6 \\
& y_1, y_2, y_3 \geq 0
\end{aligned}
$$

$$y^* = (0,5,1)^T$$
$$g(y^*) = 1900$$

Duality in LP

- **Weak Duality:** if x is feasible for the Primal problem and y is feasible for the Dual problem, then $c^T x \leq b^T y$.
  - That is, any feasible solution of the Dual problem is an Upper Bound of the Primal problem. The best (smaller) Upper Bound is the optimal solution of the Dual problem
  - **Proof**: $b^T y = y^T b \geq y^T A x = x^T A^T y \geq c^T y$ Hence: $b^T y \geq c^T x$
- **Strong Duality:** if the Primal problem has an <u>optimal solution</u> x^* with a finite optimal value ($f(x^*) < \infty$), then the Dual problem also has an <u>optimal value</u> $\boldsymbol{y}^*$ with a finite optimal value ($g(y^*) < \infty$), and $c^T x^* = b^T y^*$
- **Duality gap:** $\mathrm{d} = b^T y^* - c^T x^*$.

Primal problem

$$\begin{aligned} max \quad & f(x) = c^T x \\ s.t. \quad & Ax \leq b \\ & \mathrm{x} \geq 0 \end{aligned}$$

Dual problem

$$\begin{aligned} min \quad & g(y) = b^T y \\ s.t. \quad & A^T y \geq c \\ & \mathrm{y} \geq 0 \end{aligned}$$

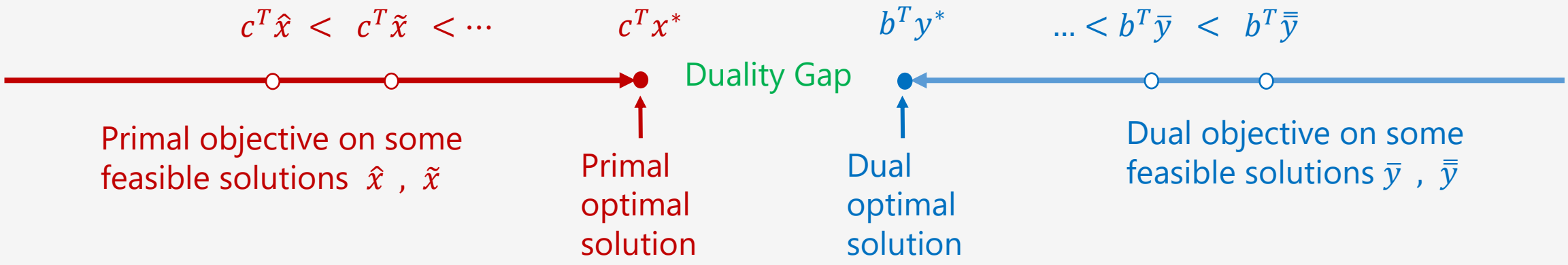
Graphical representation of Primal & Dual problems

Primal problem

$$\begin{aligned} max \quad & f(x) = c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

Dual problem

$$\begin{aligned} min \quad & g(y) = b^T y \\ s.t. \quad & A^T y \ge c \\ & y \ge 0 \end{aligned}$$

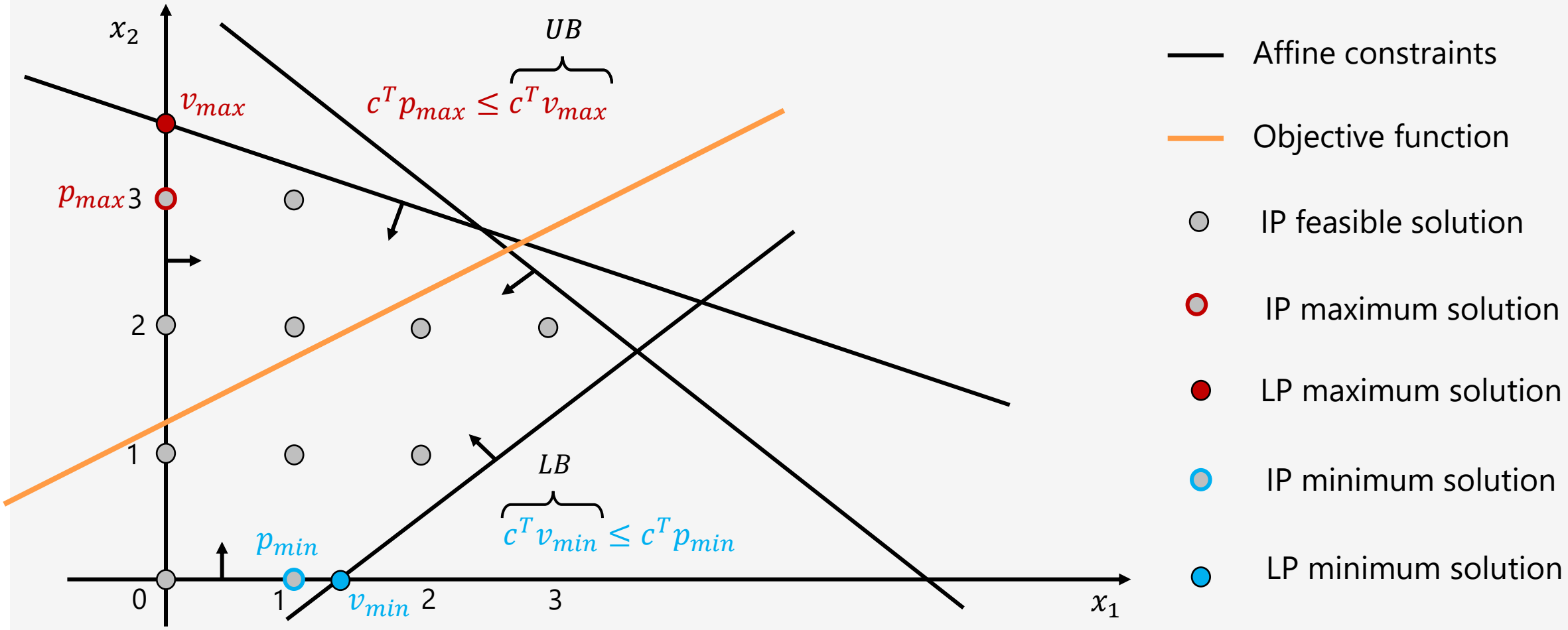


Weak duality: $c^T\tilde{x} < b^T\bar{y}$

Strong duality: $c^T x^* = b^T y^*$

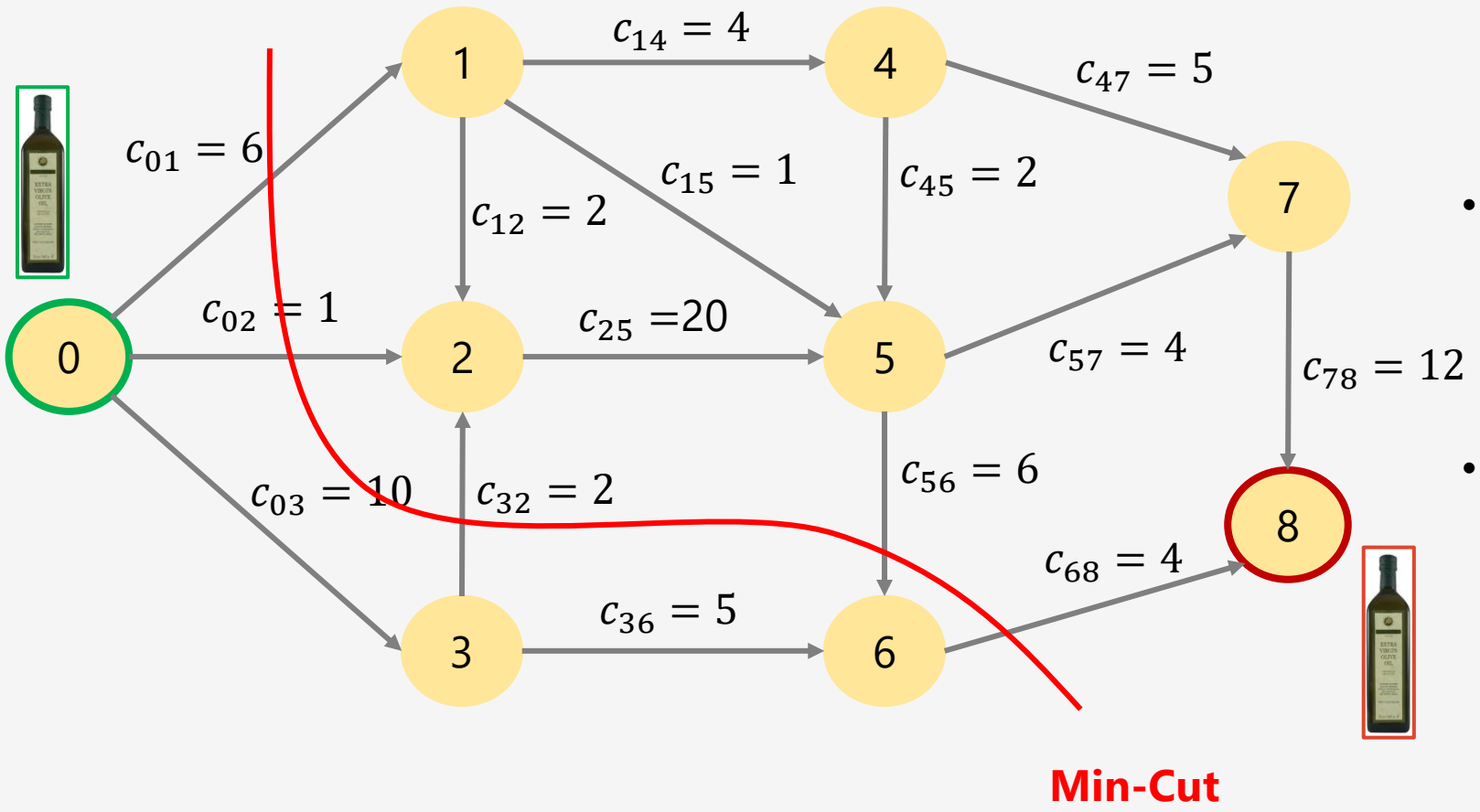
Duality theory helps us **certifying optimality** of the primal problem or to find an **upper bound** of it.

Lower and Upper Bounds based on **LP relaxations** of **IP problems**



		PRIMAL		
		Finite optimal	Unbounded	Infeasible
DUAL	Finite optimal	Possible	Impossible	Impossible
	Unbounded	Impossible	Impossible	Possible
	Infeasible	Impossible	Possible	Possible

Max-Flow and Min-Cut are dual of each other



- **Decision variables:** quantity of olive oil transported from node i to node j.
  - x_{ij}
- **Objective**: Ship as much olive oil as possible from the source node 0 to the terminal node 8.
  - $\max f(x) = x_{01} + x_{02} + x_{03}$
- **Constraints:**
  - Cannot exceed the capacity of the edges
    - $0 \leq x_{ij} \leq c_{ij}$
  - flowIN == flowOUT
    - $x_{01} = x_{14} + x_{15} + x_{12}$
    - $x_{14} = x_{47} + x_{45}$
    - $x_{15} + x_{45} + x_{25} = x_{57} + x_{56}$
    - …